Medical delivery quadcopter - design calculations

Configuration: Tarot 5008 340KV motors, 18 in props, Hobbywing XRotor-40A ESCs, Holybro PM07, 6S 18 Ah LiPo, Pixhawk 6C. Method: S.K.E.T.C.H Club formula sheet (Sections 1, 4, 5, 7), standard atmosphere constants. Date: 10 October 2026.

1. Requirements

Requirement	Target
Takeoff mass including 1 kg medkit	7.0 kg or less
Flight time on the battery	20 min or more
Range	3 km one way (medkit released at the destination, then return)
Wind	5 m/s
Thrust-to-weight ratio	2.0 or more (design target)
Energy reserve at landing	20% or more of usable energy

2. Inputs and assumptions

Quantity	Value	Basis
Gravity g; air density rho	9.81 m/s^2; 1.225 kg/m^3	Formula sheet, ISA sea level
Prop diameter D; disk area A	0.4572 m (18 in); 0.1642 m^2 per rotor	$A = \pi D^2 / 4$
Figure of merit FM	0.65	Sheet range 0.5 to 0.7
Motor efficiency; ESC efficiency	0.85; 0.95	Assumed typical
Avionics load; safety margin	25 W; +15% on propulsion power	Assumed
Drag area Cd x A	0.09 m^2	Assumed (frame, battery, payload)
Thrust coefficients C_T, C_P	0.10, 0.04	Assumed typical for 18 x 5.5 prop
Motor	Tarot 5008, Kv 340, max thrust 3.7 kgf	Tarot data
Battery	6S1P 22.2 V, 18 Ah = 399.6 Wh; mass 2.10 kg	Mass interpolated from Tattu 6S packs
Usable energy	399.6 x 0.80 x 0.95 = 303.7 Wh	Depth of discharge 80%, high-rate derate 0.95
Loaded pack voltage	21.5 V	Assumed, for current estimates
Takeoff mass M_0; mass after release M_1	6.38 kg; 5.38 kg	Component list (dry 5.38 kg + 1 kg medkit)

3. Mass and thrust margin

Hover thrust per motor: $T = M g / 4$. Thrust-to-weight: $TW = 4 T_{max} / (M g)$.

Quantity	Substitution	Result
Takeoff mass	5.38 kg + 1.00 kg	6.38 kg, margin 0.62 kg to 7 kg **PASS**
Hover thrust per motor	6.38 x 9.81 / 4	15.65 N (1.59 kgf)
Maximum thrust per motor	3.7 kgf x 9.81	36.3 N
Hover as fraction of maximum	1.59 / 3.7	43%
Thrust-to-weight, rated	4 x 3.7 / 6.38	2.32 **PASS**
Thrust-to-weight, derated 90%	0.9 x 4 x 3.7 / 6.38	2.09 **PASS**

4. Hover power (momentum theory, sheet Section 5)

$P_{ideal} = T^{1.5} / sqrt(2\ rho\ A)$. $P_{shaft} = P_{ideal} / FM$. $P_{el} = P_{shaft} / (eta_{motor}\ eta_{ESC})$. Check with the sheet form: $T = (2\ rho\ A\ (FM\ P_{shaft})^2)^{1/3}$.

Per motor unless stated	Loaded (6.38 kg)	After release (5.38 kg)
Thrust T	15.65 N	13.19 N
Ideal power	97.6 W	75.6 W
Shaft power (FM 0.65)	150.1 W	116.2 W
Electrical power (efficiency 0.81)	185.9 W	144.0 W
Total, four motors + 25 W avionics, nominal	769 W	601 W
Total with +15% margin (design value)	880 W	687 W
Overall hover efficiency eta_{tot} = FM x 0.85 x 0.95	0.525	0.525

Prop size check: a 16 in prop would need 209 W per motor against 186 W for 18 in (12.5% more), so 18 in is kept.

5. Propeller coefficient cross-check (sheet Section 5)

$T = C_T\ rho\ n^2\ D^4$. $P = C_P\ rho\ n^3\ D^5$. $N_{no\text{-}load} = Kv \times V$. $J = V_{axial} / (n\ D)$.

Quantity	Result	Comment
Hover rotor speed n	54.1 rev/s = 3244 rpm	From T and C_T
Power from C_P	154.7 W	Within 3% of momentum result (150.1 W)
No-load speed at 22.2 V	340 x 22.2 = 7548 rpm	Hover is 43% of it, matching the hover thrust fraction
Advance ratio at 16 m/s (axial part only)	J = 0.142	Well below zero-thrust value of about 0.31 (pitch / diameter) **PASS**

6. Currents and component limits (sheet Section 5)

$I = P_{el} / V$. I_{max} = C-rate x capacity. Peak case: all four motors at 3.7 kgf, P_{el} per motor = 657 W.

Check	Value	Limit	Result
Hover current, whole aircraft	41 A (2.3C)	Pack 15C = 270 A	**PASS**
Cruise current at 16 m/s	57 A	PM07 90 A continuous (64%)	**PASS**
Peak current per motor	30.6 A	ESC 40 A continuous, 60 A burst	**PASS**
Peak current, all four motors	122 A (6.8C)	PM07 140 A burst (87%); pack 270 A	**PASS**
PM07 current sensing range	122 A peak	120 A analog range	Reading clips briefly at peak

7. Battery sizing (sheet Section 5)

E = capacity x voltage. Usable = E x DOD x derate. $m_{batt} = E / e_{spec}$ (sheet: 150 to 220 Wh/kg).

Quantity	Substitution	Result
Pack energy	18 Ah x 22.2 V	399.6 Wh (1.44 MJ)
Usable energy	399.6 x 0.80 x 0.95	303.7 Wh
Specific energy	399.6 Wh / 2.10 kg	190 Wh/kg, inside the sheet band **PASS**
Energy needed, worst case at 16 m/s	230 Wh used / 0.76	303 Wh nominal, about 1.59 kg of pack at this density

8. Flight time (sheet Section 4)

Sheet form: t = E DOD eta_{tot} / P_r, with P_r the aerodynamic (ideal) power. Refined: t = usable energy / electrical hover power.

Case	Result	Check
Sheet formula (upper bound, no avionics or derate)	399.6 x 0.80 x 0.525 / 390 W = 25.8 min	Upper bound
Refined, loaded, nominal power	303.7 Wh / 769 W = 23.7 min	**PASS**
Refined, loaded, with +15% margin	303.7 Wh / 880 W = 20.7 min	**MARGINAL** (target 20 min, narrow)
After release, nominal / with margin	30.3 min / 26.5 min	**PASS**

9. Wind and drag (sheet Section 1)

D = 0.5 rho V^2 (Cd A). Tilt = atan(D / W). Cruise power = rotor power at total thrust sqrt(W^2 + D^2) plus D x V.

Case	Drag	Tilt	Power (with margin)
Holding position in 5 m/s wind	1.4 N	1.3 deg	about hover power
Holding position in 10 m/s wind	5.5 N	5.0 deg	about hover power
Cruise at 14 m/s, loaded	10.8 N	9.8 deg	1115 W (after release 923 W)
Cruise at 16 m/s, loaded	14.1 N	12.7 deg	1234 W (after release 1044 W)

10. Mission energy and range (sheet Section 4)

Phases: climb 1 min (1.2 x hover power, 6.38 kg), cruise out 3 km, descent 1 min (hover power), release on the ground 30 s (avionics only), climb 1 min (5.38 kg), cruise back 3 km, landing 1 min. Ground speed = airspeed minus headwind. Pass = at least 20% of usable energy left. Sheet range R = V t is the cruise-only upper bound.

Case	Time	Energy	Left	Result
Airspeed 14 m/s				
No wind	11.6 min	179 Wh	41%	**PASS**
Crosswind 5 m/s	12.1 min	188 Wh	38%	**PASS**
Headwind out, tailwind back (stated)	12.7 min	201 Wh	34%	**PASS**
Headwind both ways (worst case)	15.6 min	246 Wh	19%	**FAIL**
Airspeed 16 m/s				
No wind	10.8 min	176 Wh	42%	**PASS**
Crosswind 5 m/s	11.1 min	183 Wh	40%	**PASS**
Headwind out, tailwind back (stated)	11.4 min	193 Wh	37%	**PASS**
Headwind both ways (worst case)	13.6 min	230 Wh	24%	**PASS**

Quantity	14 m/s	16 m/s
Maximum one-way distance, worst case (20% reserve)	2.95 km	3.22 km
Maximum one-way distance, no wind	4.58 km	4.68 km
Still-air range R = V t (cruise only)	13.7 km	14.2 km

Result: at 14 m/s the worst case falls just under the 20% reserve (2.95 km one way). At **16 m/s** every case passes, and the worst case reaches 3.22 km. Design cruise speed: 16 m/s airspeed, with the aircraft parachute kept. Without the parachute (-0.5 kg) 14 m/s would also pass (27% left, 3.36 km).

11. Arm strength (sheet Section 7)

$I = \pi (d_o^4 - d_i^4) / 64$. $\sigma = M y / I$. $\delta = P L^3 / (3 E I)$. $FS = \sigma_{yield} / \sigma_{max}$. Arm: 25 x 23 mm carbon tube, L = 0.35 m, E = 70 GPa and σ_{yield} = 500 MPa (assumed, confirm with supplier).

Quantity	Substitution	Result
Second moment of area I	$\pi (0.025^4 - 0.023^4) / 64$	5.44e-09 m^4
Root bending moment (max thrust, load factor 2)	2 x 36.3 N x 0.35 m	25.4 N m
Bending stress	25.4 x 0.0125 / 5.44e-09	58 MPa
Factor of safety	500 / 58	8.6 (sheet: 1.5 to 2.0) **PASS**
Tip deflection at hover thrust	$15.6 \times 0.35^3 / (3 \times 70e9 \times I)$	0.59 mm
Tip deflection at max thrust	$36.3 \times 0.35^3 / (3 \times 70e9 \times I)$	1.36 mm

12. Centre of gravity (sheet Section 7)

$x_{cg} = \sum(m_i x_i) / m_{total}$. Mass shift: $\delta x_{cg} = m_{shifted} \delta x / m_{total}$. Layout (forward positive, mm): battery, medkit and servo at 0; parachute +30; Pixhawk +20; strobe +100; PM07 -40; BEC -60; fuse -90; RC receiver -100; telemetry -100; GPS mast -120; wires -30. Motors and frame are symmetric.

Quantity	Result
CG position, loaded	x = 0.0 mm, y = 0.0 mm **PASS**
CG shift when the medkit is released, payload on the CG axis	0.0 mm
Same, payload offset 10 mm	1.6 mm
Same, payload offset 20 mm	3.1 mm
Design rule	Mount the payload within 10 mm of the CG axis

13. Wire losses

R = resistance per metre x length. Loss = $I^2 R$. 8 AWG = 2.06 mOhm/m, 14 AWG = 8.28 mOhm/m.

Run	Resistance	At peak	At hover
8 AWG main lead, 0.3 m round trip	0.62 mOhm	9.2 W at 122 A	1.0 W at 41 A
14 AWG ESC feed, 0.4 m round trip	3.3 mOhm	3.1 W at 31 A	0.3 W at 10 A

Losses are small, so both gauges are acceptable.

14. Command link budget (assumed radio parameters)

Free-space path loss = $20 \log_{10}(d \text{ in km}) + 20 \log_{10}(f \text{ in MHz}) + 32.44$. Margin = $P_{tx} + G_{tx} + G_{rx}$ - cable loss - path loss - receiver sensitivity.

Quantity	Value
Path loss at 3 km, 868 MHz	100.8 dB
Assumed: transmit 27 dBm, antennas 2 dBi each, cable loss 1 dB, receiver sensitivity -105 dBm	Link margin 34.2 dB **PASS**

15. Requirement check

Requirement	Result	Status
Takeoff mass 7 kg or less	6.38 kg	**PASS**
Thrust-to-weight 2.0 or more	2.32 rated, 2.09 derated	**PASS**
Flight time 20 min or more	20.7 min hover with margin, 23.7 min nominal	**MARGINAL**
3 km one way in 5 m/s wind	Stated wind: 37% energy left at 16 m/s. Worst case: 24% left, 3.22 km maximum	**PASS**
ESC, pack and PM07 current limits	Hover 41 A, cruise 57 A, peak 122 A	**PASS**
Arm strength	Factor of safety 8.6	**PASS**
Centre of gravity	Within 1 mm, payload on the CG axis	**PASS**

Design decisions from these calculations: cruise at 16 m/s airspeed; keep the payload on the CG axis; the 18 Ah pack meets the 20 min flight time only narrowly (20.7 min with the 15% margin), so if a hover test shows more than about 900 W, move to a larger pack.

Assumptions to verify before submission: efficiencies and figure of merit, drag area, C_T and C_P, arm material properties, radio parameters, and the weight of the exact 18 Ah LiPo pack. Cross-check the power numbers with a multicopter calculator and the manufacturers data sheets.